English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the handwritten solutions. The source's order, calculations and unfinished evaluations are retained.

Problem 1

1. $\mathbb{P}(A)\mathbb{P}(B) = 0.12$, and

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)\\ &= \mathbb{P}(A) + \mathbb{P}(B) - (1 - \mathbb{P}(\overline{A \cup B}))\\ &= 0.7 - 0.58 = 0.12.\end{aligned}$$

   Thus $\mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A \cap B)$: yes, they are independent.

2.
$$\mathbb{P}(\text{three correct}) = \binom{5}{3}(\tfrac{1}{4})^3(\tfrac{3}{4})^2 \approx 8.79\%,$$
$$\mathbb{P}(\text{questions 1,3,5 correct} \mid \text{three correct}) = \frac{1}{\binom{5}{3}} = \frac{1}{10}.$$

3. Under independence, all the covariances are zero. In general, use the formula (or definition):

$$\begin{aligned}\operatorname{Cov}(X+Y, Z) &= \mathbb{E}[(X+Y)Z] - \mathbb{E}[X+Y]\mathbb{E}Z\\ &= \mathbb{E}[XZ] - \mathbb{E}X\mathbb{E}Z + \mathbb{E}[YZ] - \mathbb{E}Y\mathbb{E}Z\\ &= \operatorname{Cov}(X, Z) + \operatorname{Cov}(Y, Z),\end{aligned}$$

   by linearity.

4. Let $X_i = 1$ if toss i is heads and zero otherwise; the X_i are iid Ber(1/2). We seek

$$\operatorname{Cov}(X_1 + \cdots + X_7, (1 - X_4) + \cdots + (1 - X_{10})).$$

   By part 3 this is $4\operatorname{Cov}(X_i, -X_i) = -4DX_i = -1$, since $\operatorname{Cov}(X_i, X_j) = 0$ for $i \neq j$. Writing the two sums as Y, Z, the correlation is
$$\frac{-1}{\sqrt{DY\,DZ}} = \frac{-1}{7 \cdot \frac{1}{4}} = -\frac{4}{7}.$$

5. Ending after exactly three rounds means no ace among cards 1–20 and at least one ace among cards 21–30. Thus

$$\begin{aligned}\mathbb{P}(\text{exactly three rounds}) &= \mathbb{P}(\text{no ace in cards 1–20}) - \mathbb{P}(\text{no ace in cards 1–30})\\ &= \frac{\binom{48}{20}}{\binom{52}{20}} - \frac{\binom{48}{30}}{\binom{52}{30}}\\ &= \frac{48!}{20!28!}\frac{20!32!}{52!} - \frac{48!}{30!18!}\frac{30!22!}{52!}\\ &= \frac{29 \cdot 30 \cdot 31 \cdot 32}{49 \cdot 50 \cdot 51 \cdot 52} - \frac{19 \cdot 20 \cdot 21 \cdot 22}{49 \cdot 50 \cdot 51 \cdot 52} \approx 10.58\%.\end{aligned}$$

6. Let $X_1, \ldots, X_6$ be the face labels, iid uniform on $\{1, \ldots, 6\}$, and let Y be the roll. Then

$$\begin{aligned}\mathbb{P}(\text{six}) &= \mathbb{E}[\mathbb{1}_{\{\text{six}\}}]\\ &= \mathbb{E}\left[\mathbb{E}\left[\frac{1}{6}\mathbb{1}_{\{X_1=6\}} + \cdots + \frac{1}{6}\mathbb{1}_{\{X_6=6\}} \;\middle|\; X_1, \ldots, X_6\right]\right]\\ &= \mathbb{E}[\mathbb{1}_{\{X_i=6\}}] = \frac{1}{6}.\end{aligned}$$

   Alternatively, condition on the number of sixes, which ranges from 0 to 6. The expected number of rolls until the product is divisible by 6 is ∞ (or one may say it does not exist). The die could, for example, have only ones; then the product can never be divisible by 6.

Problem 3

1. Coupon collector: see the exercises.

$$\text{Answer: } 10(1+\tfrac{1}{2}+\cdots+\tfrac{1}{10}) \approx 29.29.$$

2. $X_i \sim \mathrm{Ge}(1/10)$ and $Y = \max\{X_1, \ldots, X_{10}\}$.

3.

$$\max\{x_1,\ldots,x_n\} = \sum_i x_i - \sum_{i<j} \min\{x_i,x_j\} + \sum_{i<j<k} \min\{x_i,x_j,x_k\} - \cdots + (-1)^{n+1} \min\{x_1,\ldots,x_n\}.$$

4. $\mathrm{Ge}(p_1+p_2-p_1p_2)$. See the exercises.

5. For four species with probabilities $1/10, 2/10, 3/10, 4/10$,

$$\min\{X_i, X_j\} \sim \mathrm{Ge}(p_i+p_j).$$

This does not follow from part 4, because the X_i, X_j are not independent; it follows by another argument (what is it?). Hence

$$\begin{aligned}
\mathbb{E}[\max\{X_1,\ldots,X_4\}] &= \mathbb{E}[\sum_i X_i] - \mathbb{E}[\sum_{i<j} \min\{X_i,X_j\}] + \ldots \\
&= \left[10 + \frac{10}{2} + \frac{10}{3} + \frac{10}{4}\right] \\
&\quad - \left[\frac{10}{3} + \frac{10}{4} + \frac{10}{5} + \frac{10}{5} + \frac{10}{6} + \frac{10}{7}\right] \\
&\quad + \left[\frac{10}{6} + \frac{10}{7} + \frac{10}{8} + \frac{10}{9}\right] - 1 = \ldots
\end{aligned}$$

Problem 2

1. Let A_i mean that player i receives BGN 0. Then

$$\begin{aligned}
\mathbb{P}(A_1 \cup A_2 \cup A_3) &= \mathbb{P}(\text{at least one receives 0}) \\
&= \sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) + \mathbb{P}(A_1 \cap A_2 \cap A_3) \\
&= 3\mathbb{P}(A_1) - 3\mathbb{P}(A_1 \cap A_2) + \mathbb{P}(A_1 \cap A_2 \cap A_3) \\
&= 3\sum_{\ell=0}^{\infty} (\tfrac{1}{2})^{\ell+1} \left[(\tfrac{2}{3})^\ell - (\tfrac{1}{3})^\ell\right] + \frac{1}{2} \\
&= \frac{3}{2}\left[\frac{1}{1-\frac{1}{3}} - \frac{1}{1-\frac{1}{6}}\right] + \frac{1}{2} \\
&= \frac{3}{2}\left[\frac{3}{2} - \frac{6}{5}\right] + \frac{1}{2} = \frac{19}{20}.
\end{aligned}$$

The answer is $1 - \mathbb{P}(\bigcup_i A_i) = 1/20 = 5\%$.

2. Let X be the winnings of a fixed player. For $k \geq 0$,

$$\mathbb{P}(X=k) = \sum_{\ell=0}^{\infty} (\tfrac{1}{2})^{\ell+1} \binom{\ell}{k} (\tfrac{1}{3})^k (\tfrac{2}{3})^{\ell-k}.$$

Here $(1/2)^{\ell+1}$ is the probability that the casino pays BGN ℓ, and the remaining factor is the probability that k of these payments go to the fixed player. Put $\ell = k+m$:

$$\begin{aligned}
\mathbb{P}(X=k) &= \sum_{m=0}^{\infty} (\tfrac{1}{2})^{k+m+1} \binom{k+m}{k} (\tfrac{1}{3})^k (\tfrac{2}{3})^m \\
&= (\tfrac{1}{6})^k \frac{1}{2} \sum_{m=0}^{\infty} (\tfrac{1}{3})^m \binom{k+m}{k}
\end{aligned}$$

$$= (\tfrac{1}{6})^k \frac{1}{2} \frac{1}{(1-\frac{1}{3})^{k+1}} = \frac{1}{2}(\tfrac{1}{6})^k (\tfrac{3}{2})^{k+1} = \frac{3}{4}(\tfrac{1}{4})^k.$$

The sum uses the formula supplied in the test. The handwritten conclusion is

$$X \sim \mathrm{Ge}(\tfrac{1}{4}) \quad \Longrightarrow \quad DX = \frac{3/4}{(1/4)^2} = 12.$$